Decoder-Free Optical Learning for Axial Structure Recovery in Fixed-Focus Fluorescence Microscopy

Supplementary Material

This supplement specifies the optical model, plate training, reconstruction and evaluation protocols. It reports the full numerical comparisons and complementary structure and uncertainty analyses. Section, equation and figure numbers without an S prefix refer to the main paper. Unless stated otherwise, additional experiments use the 12 prespecified neuronal crops, with five direction subsets at $C > 1$.

S1. Optical implementation and discretization

Coordinates and angular scan. The optical axis is z and $\mathbf{r} = (x, y)$ is transverse position. In the simulated input medium, $n = 1$ and $k_{\mathrm{ex}} = 2\pi/\lambda_{\mathrm{ex}}$. The incident direction is

$$\mathbf{k}_t/k_{\mathrm{ex}} = (\cos\theta_t \cos\alpha_t, \cos\theta_t \sin\alpha_t, \sin\theta_t). \tag{S1}$$

Thus, θ is the implemented vertical-plane parameter, not a polar angle from the optical axis. We use $\alpha_m = m\pi/16$, $m = 0, \ldots, 15$, with eight equally spaced θ values in each of $[\pi/6, \pi/3]$ and $[2\pi/3, 5\pi/6]$. The second interval changes the sign of the transverse component while retaining positive longitudinal propagation. The resulting 256 directions are distinct. For $C = 4$ and 8, five fixed random subsets of 64 and 32 directions are shared across encoders.

Propagation and sampling. The incident field is uniformly x-polarized before phase modulation. Angular-spectrum propagation over $D = 6.6\,\mu\mathrm{m}$ carries the field from the plate to the intermediate illumination image plane. A finite NA-0.8 pupil filters the modulated angular spectrum before a vectorial Richards–Wolf relay generates three specimen-field components [7, 9]. Incident direction and transmitted angular spectrum are different quantities; the input direction range need not lie entirely inside the relay pupil. We sum the component intensities. Detection uses an NA-0.1 emission intensity PSF. Excitation and emission share 532 nm in this simulation; a physical implementation would use the actual spectral bands. No scattering, aberration or fluorescence saturation is modeled.

The specimen field is sampled laterally at $0.08\,\mu\mathrm{m}$ and axially at $0.32\,\mu\mathrm{m}$. Lateral binning gives an isotropic $0.32\,\mu\mathrm{m}$ analysis grid of $32 \times 96 \times 96$ cells. The reconstruction basis averages $2 \times 2 \times 2$ analysis cells into each $0.64\,\mu\mathrm{m}$ voxel. Camera pixels average 8×8 analysis samples laterally, giving $2.56\,\mu\mathrm{m}$ pixel pitch and 12×12 samples per image. Distances and pitches are expressed in specimen-equivalent coordinates.

Aperture and control encoders. The phase plate has $0.32\,\mu\mathrm{m}$ features and initially independent phases uniform on $[0, 2\pi)$. It is simulated on an aperture twice the specimen width, then the excitation is cropped to the specimen. Only the central plate region is optimized; its independent random surround remains fixed. The defocused amplitude controls use binary masks with the same feature pitch and 50 % open pixels at the same plate plane. The in-focus DMD uses independently drawn binary patterns relayed into focus. These are controls for the tested fixed-focus acquisition, not a comparison against all DMD microscopy configurations.

A finite illumination edge can itself encode depth when tilted. The plate-free control therefore uses an aperture four times the specimen width. Its residual central/border angular modulation is about 1.0/1.2 %, compared with 5.6/37 % at twice the width. The $4\times$-aperture plate-free study is kept separate from the $2\times$ phase/mask comparison. The random plate changes little across these apertures (centroid error 0.446 versus $0.431\,\mu\mathrm{m}$ at $C = 8$, pretrained PnP), but that does not make their learned-versus-plate-free contrast a single-aperture experiment.

Integrated operator. The detection footprint $g_{p,i}$ includes the emission PSF, camera-pixel average and fine-cell integration weight. Reusing it in $A_{(t,p),j} = \sum_{i \in \mathcal{P}_j} I_t(\mathbf{r}_i, z_i) g_{p,i}$ avoids propagating each reconstruction basis element separately. Equality $\mathsf{A} = \mathsf{BU}$ is an algebraic statement for the selected piecewise-constant basis. It does not assert that a real specimen is constant within a voxel. Test measurements use $\mathsf{B}\mathbf{x}_f$, while the reference for reconstruction is $\mathsf{R}\mathbf{x}_f$, retaining the residual $\mathsf{B}(\mathbf{x}_f - \mathsf{UR}\mathbf{x}_f)$.

Computational scope. After detection-footprint precomputation, direct assembly costs $O(TPF)$, compared with $O(TNF \log F)$ for constructing columns through separate FFT-based forward simulations: T illumination patterns, P camera samples, F fine-grid cells and N reconstruction voxels. Representative timings on an RTX 4500 Ada are approximately 20 minutes for 1000 optical-training steps, six minutes to assemble a full plate operator, 4–6 seconds for a PnP reconstruction at $C = 8$, and 17 seconds for TV.

S2. Plate training and objective comparisons

Signatures and sampling. For each lateral position and depth, we collect the detector-integrated response of its reconstruction voxel over the sampled illumination directions

and camera pixels. Four directions and 24 interior lateral positions are sampled per Adam step. Each full-field depth signature therefore has 4×144 entries during training. Its mean over all angle–pixel entries is removed before unit-norm normalization. This is not per-pixel temporal centering. The main paper's evaluation statistic uses the same centering definition over the 32 directions of draw 0 and the 44×44 interior lateral positions (indices 2–45). It averages absolute off-diagonal depth correlations, not the squared weighted training loss.

With zero-based reconstruction-plane indices $i, j \in \{0, \dots, 15\}$, off-diagonal weights are

$$w_{ij} = \begin{cases} 2, & i + j = 14, \\ 1, & i + j \neq 14, \end{cases} \qquad i \neq j. \tag{S2}$$

This emphasizes reflected pairs around grid plane 7. It is an implemented weighting choice, not a universal symmetry of a phase screen or a statement that plane 7 is exactly the physical focus. Both orderings of each pair are included and the sum is divided by the total weight. Norm denominators include 10^{-12} for numerical stability.

We use 1000 Adam steps with learning rate 0.05. The two training seeds change the sampled positions and directions, starting from the same phase realization and frozen surround. No specimen examples, reconstruction loss or denoiser enter this process. Normalizing a signature removes uniform power scaling from the loss; dose normalization is applied separately when generating evaluation measurements.

Why angular modulation matters. Consider a nonnegative temporal response $\mathbf{e}_z$ at one camera pixel, with mean $\mu_z > 0$. Write $\mathbf{e}_z = \mu_z \mathbf{1} + \tilde{\mathbf{e}}_z$, and define angular contrast $\kappa_z = \|\tilde{\mathbf{e}}_z\|/(\sqrt{T}\mu_z)$. If ρ is the correlation of the mean-removed fluctuations, the raw cosine is exactly

$$c_{zz'} = \frac{1 + \rho\kappa_z\kappa_{z'}}{\sqrt{(1+\kappa_z^2)(1+\kappa_{z'}^2)}}. \tag{S3}$$

The shared mean dominates when contrasts are small, even if the fluctuations are weakly correlated. Increasing contrast and placing strong responses at different angles can reduce this overlap. At the preset field center, median contrast is 0.58/1.53 and mean raw cosine 0.83/0.41 for random/learned plates. Only one preset position is used for this illustrative decomposition; it is not a field-wide causal ablation.

A common-footprint approximation explains why centering the full angle–pixel signature does not necessarily remove the shared angular component. If $\mathbf{q}_z = \mathbf{e}_z \otimes \mathbf{g}$ with a fixed P-pixel footprint, global centering gives correlation

$$\frac{1 - \eta + \rho\kappa_z\kappa_{z'}}{\sqrt{(1-\eta+\kappa_z^2)(1-\eta+\kappa_{z'}^2)}}, \quad \eta = \frac{(\mathbf{1}^\top \mathbf{g})^2}{P\|\mathbf{g}\|^2}. \tag{S4}$$

For a localized footprint, η can be small, so a common angular component remains. The actual operator includes depth-varying footprints and within-voxel variation; neither training nor evaluation uses this separable approximation. This provides an interpretation of the observed code, not a proof that contrast alone causes the reconstruction gain or that the objective is universally optimal.

Worst-case coherence surrogate. Let ρ_{ij} be the inner product of two centered, unit-norm depth signatures. The comparison uses

$$\mathcal{L}_{\text{coh}} = \mathbb{E}_{\mathbf{r},\mathcal{T}} \left[\frac{1}{50} \log \sum_{i \neq j} \exp(50|\rho_{ij}|) \right]. \tag{S5}$$

This smooth maximum targets the worst pair rather than the weighted average of squared correlations. It is motivated by coherence-based sensing and PSF design [3, 4, 13]; it does not implement Miniscope3D's full optical design or lateral-shift cross-coherence objective.

Depth Fisher-information surrogate. Let R contain the raw signatures in a sampled batch. The implementation uses $s = R/\operatorname{stopgrad}(\max(\operatorname{mean} R, 10^{-30})) + 10^{-3}$. Finite differences s'_z are taken over depth-plane index, centrally inside the grid and one-sided at its ends. With brightness treated as a nuisance parameter, the depth-information surrogate is

$$\mathcal{I}_z = \sum_m \frac{(s'_{z,m})^2}{s_{z,m}} - \frac{(\sum_m s'_{z,m})^2}{\sum_m s_{z,m}}, \tag{S6}$$

where m indexes angle–pixel entries. Training minimizes $-\operatorname{mean}_z \log \max(\mathcal{I}_z, 10^{-12})$. This uses the Schur-complement form of Poisson Fisher information [10], with the stated normalization and discretization. It is an implemented surrogate, not a comparison with every Fisher-based design.

Matched budget and centering ablation. All objectives use the same initial phase, sampled positions/directions, optimizer, learning rate and step count. Reconstruction parameters are validated independently for each resulting operator. These conditions match computational budgets, not convergence rates or optimal hyperparameters for every criterion. The uncentered variant omits mean removal and retains the weights and squared-correlation form. Under neuron-trained PnP at $C = 8$, it is essentially tied with the centered version (W1 0.207 versus 0.206 μm). Centering is therefore not an established explanation of the gain over the tested alternatives.

Comparator	Seed	Reconstruction	W1, other	W1, ours	W1 change [95 % CI]	Centroid change [95 % CI]
Smooth worst-case	0	TV	0.238	0.196	−17 % [−20, −15]	−18 % [−21, −15]
Smooth worst-case	0	Pretrained PnP	0.346	0.277	−20 % [−25, −16]	−22 % [−28, −17]
Smooth worst-case	0	Neuron-trained PnP	0.238	0.206	−13 % [−15, −12]	−12 % [−16, −9]
Smooth worst-case	1	TV	0.233	0.195	−16 % [−19, −14]	−16 % [−19, −13]
Smooth worst-case	1	Pretrained PnP	0.338	0.279	−17 % [−23, −13]	−18 % [−25, −12]
Smooth worst-case	1	Neuron-trained PnP	0.229	0.200	−13 % [−15, −11]	−12 % [−16, −10]
Depth Fisher surrogate	0	TV	0.228	0.196	−14 % [−16, −12]	−15 % [−17, −11]
Depth Fisher surrogate	0	Pretrained PnP	0.340	0.277	−19 % [−22, −17]	−20 % [−24, −17]
Depth Fisher surrogate	0	Neuron-trained PnP	0.240	0.206	−14 % [−17, −12]	−14 % [−19, −10]
Depth Fisher surrogate	1	TV	0.227	0.195	−14 % [−16, −12]	−15 % [−16, −13]
Depth Fisher surrogate	1	Pretrained PnP	0.332	0.279	−16 % [−20, −12]	−16 % [−23, −11]
Depth Fisher surrogate	1	Neuron-trained PnP	0.239	0.200	−16 % [−18, −14]	−17 % [−18, −15]

Table S1. Objective comparisons on the 12 prespecified crops at $C = 8$. Errors are in μm; changes compare our plate with the comparator of the same training seed, with paired cell-bootstrap 95 % intervals. W1 and centroid are computed from the same completed reconstruction rows. Negative changes favor our criterion.

S3. Reconstruction and validation

Shared framework. The encoder changes the forward operator; the prior and solver family remain fixed across encoders. TV uses $\frac{1}{2}\|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \lambda\|\nabla\mathbf{x}\|_{2,1}$ with a box constraint. PnP uses the quadratic data update and denoising/sparsity branches described in the main paper. A quadratic data term penalizes residuals but does not guarantee membership in the hard measurement-interval set used in Sec. S8. Generic denoisers need not minimize an explicit regularization energy.

TV. We use isotropic 3D forward-difference TV and ADMM with separate gradient and box branches, both with unit penalty. The box branch projects its auxiliary variable at each iteration, and the final volume is clipped. There are 300 ADMM iterations; each conjugate-gradient volume update uses at most 20 iterations at relative tolerance 10^{-5}. The starting volume is normalized back-projection clipped to $[0, 1]$. The regularization weight is selected from $\{0.01, 0.03, 0.1, 0.3, 1, 3\}$ using mean validation PSNR. Edge selections extend the grid by a factor of three, at most twice. These fixed iteration budgets are not claims of exact minimization. Additional checks with 1000 iterations changed mean PSNR by -0.03 dB at $C = 8$ and centroid error by at most $0.001\,\mu$m; checks at $C = 1$ changed PSNR by at most 0.03 dB and centroid error by $0.006\,\mu$m.

PnP. Both variants run 20 ADMM iterations, with at most 20 conjugate-gradient steps per volume update and relative tolerance 10^{-5}. Denoising-branch penalties are $5s$ and the sparsity penalty is $10s$. The implementation sets nominal branch weights of 0.2 for the slice denoisers and 1.0 for the volumetric denoiser. The fixed denoisers ignore this proximal-scale argument, so these values activate the branches rather than adjusting denoising strength. Their coupling to the data is controlled by the shared penalty. The sparsity weight is $0.2l$; branches with zero weight are omitted. Validation selects $l = 0$ for all PnP settings in the main 60-crop comparison; sparsity is active only in some supplementary conditions. We validate $s \in \{2^{-4}, 2^{-3}, \ldots, 4\}$ and $l \in \{0, 1/16\}$. Boundary selections extend the scale grid by a factor of two and the nonzero sparsity scale by four, at most twice. Final selections are interior. All PnP results start from normalized back-projection and are clipped to $[0, 1]$ only after the final iteration.

Denoisers. Pretrained PnP uses SCICO's 17M natural-image DnCNN (17 layers, trained at noise standard deviation 0.1) [2, 14], independently along three slice orientations. Its denoising noise level is fixed; it does not receive a specimen-dependent noise estimate. Neuron-trained PnP replaces this 2D model and adds a 3D branch. The 2D network has 17 layers and 64 channels; the 3D network has 12 layers and 48 channels. We do not evaluate 2D-only and 3D-only neuron-trained variants, so the comparison changes both the training distribution and denoising architecture.

Training uses 459 crops from cell 1, stacks 8 and 9 of animal SOM009, with stack 10 held out for monitoring. Duplicated planes and planes whose mean falls below half their neighbors' median are excluded before crop extraction. Both networks use Adam at 10^{-3}, Gaussian noise $\sigma \sim \mathcal{U}(0, 0.25)$, flips, lateral transpositions and intensity scaling in $[0.3, 1]$. The 2D model uses random slices for 30,000 steps at batch size 64; the 3D model uses volumes for 15,000 steps at batch size 8. Neither contributes test data or is updated during optical training or reconstruction.

Parameter selection. The two vascular validation crops come from another animal and are used for all neuronal

comparisons. Parameters are selected separately by encoder, compression and changed noise condition, with the same selection rule and fixed prior weights. No neuronal test crop is used for tuning. Optical design uses neither validation nor test specimens. Reference arrays are used only for metric reporting, not in the reconstruction updates.

S4. Data, metrics and paired statistics

Dataset partitions. The neuronal references are public two-photon stacks used in DEEP^2 [12]. Test cells are SOM004 cells 1–2, SOM005 cell 1 and SOM006 cells 1–3: six cells from three animals. Each cell contributes two prespecified crops selected by projected foreground area times depth spread. After fixing reconstruction parameters, eight additional eligible crops per cell were sampled uniformly by location, excluding the two already selected. The resulting 48 crops were reconstructed without retuning. There are 60 test crops in the main comparison; other full-field studies use the 12 prespecified crops unless stated otherwise.

Denoisers are trained on SOM009, a fourth animal, as specified in Sec. S3. Parameter selection uses two vascular crops from a different animal. Conformal calibration uses a held-out stack of SOM009 and does not tune the reconstruction. No test volume enters optical training, denoiser training, scalar tuning or interval calibration.

Profile and centroid errors. Let $g_c = \sum_z x_{\mathrm{gt}}(c, z)$ and retain columns with $g_c > 0.1 \max_{c'} g_{c'}$. We set $\pi_c = g_c / \sum_{c'} g_{c'}$ over retained columns and normalize each nonnegative profile to unit mass. The one-dimensional W1 distance equals Δz times the sum of absolute CDF differences; the main paper averages it with π_c. All included reconstructed columns have positive mass in the reported evaluations. The centroid error uses exactly the same columns and weights. For normalized profiles, W1 is at least the absolute centroid difference. A centroid can match when the profile does not, but similar reductions in the two errors do not establish a larger gain in profile shape.

PSNR uses the complete volume and SCICO's default reference range, $\max(\mathbf{x}_{\mathrm{gt}}) - \min(\mathbf{x}_{\mathrm{gt}})$ [2]. Foreground SSIM uses a Gaussian-weighted 3D SSIM map ($\sigma = 1.5$, intensity range one), averaged over reference voxels above 10 % of peak intensity after two binary-dilation steps. These intensity metrics complement W1, which normalizes each column and therefore does not penalize brightness errors by itself.

Depth-blind references. The flat estimate preserves each column's total brightness but distributes it uniformly over depth. The depth-blind oracle preserves that brightness and uses the reference crop's global axial profile for every column. The latter has access to the correct projected structure and global depth distribution but no local depth placement. On the 12 prespecified crops, their W1 errors are 2.266 and 1.432 μm, with centroid errors 2.148 and 1.322 μm, respectively. They are context for interpreting axial metrics, not reconstruction methods supplied with the actual measurements.

Pairing and uncertainty. For each setting, encoders share the specimen, direction subset, nominal dose and noise-generation protocol. We average the five draws within each crop. A relative change is the ratio of learned and comparator crop means minus one, not the mean of per-crop percentages. Confidence intervals use 10,000 cell-bootstrap resamples, keeping all crops of a sampled cell together. Cells from the same animal may also be dependent; six cells from three animals do not support a precise population-level guarantee. The intervals describe this finite evaluation and are not corrected for all exploratory comparisons.

Consistent reconstruction metrics. W1, centroid, PSNR and foreground SSIM in Tab. S2 are computed from the same completed reconstruction rows. Each of the nine configurations contains all 60 crops and five draws. The reconstruction reruns use the recorded operators, parameters, directions and noise. Relative to earlier runs, the largest per-row differences are 0.019 dB/0.0004 μm for TV and 0.198 dB/0.0204 μm for PnP (PSNR/centroid); mean metrics agree at the reported precision. The all-column W1 evaluation used a first-row reproduction check for each configuration before completing its remaining solves. We report the resulting numerical estimates rather than assuming bitwise reproducibility across GPU runs.

S5. Full reconstruction results

Both training seeds. Table S2 gives the complete 60-crop comparison, including both optical sampling seeds. Each learned plate lowers W1 on every crop under each prior. Seed 0 improves centroid error on 59 of 60 crops for neuron-trained PnP and all crops for the other methods. Seed 1 improves centroid error on all crops. Both plates start from the same phase realization, so these results test sampling variability, not independence of initialization.

Reference-based and random crops. Table S3 separates the 12 prespecified crops from the 48 later random crops. All methods retain positive optical gains on the latter, with parameters already fixed. This addresses dependence on the initial structure-score selection, while preserving the cell and animal limitations of the dataset.

Comparison with untrained encoders. The separate centroid study in Tab. S4 uses pretrained PnP, the 12 prespecified

prior	plate	W1 (μm)	centroid (μm)	PSNR (dB)	FG-SSIM	ΔW1 [95 % CI]	Δcentroid [95 % CI]
TV	random	0.371	0.259	36.70	0.850	–	–
	learned s0	0.218	0.150	38.29	0.910	−41 % [−45, −38]	−42 % [−46, −38]
	learned s1	0.218	0.149	38.31	0.910	−41 % [−45, −38]	−42 % [−46, −38]
Pretrained PnP	random	0.579	0.490	35.72	0.806	–	–
	learned s0	0.290	0.224	37.56	0.900	−50 % [−53, −47]	−54 % [−57, −51]
	learned s1	0.289	0.222	37.58	0.900	−50 % [−53, −47]	−55 % [−58, −51]
Neuron-trained PnP	random	0.271	0.197	38.01	0.902	–	–
	learned s0	0.195	0.139	39.60	0.934	−28 % [−29, −26]	−30 % [−32, −27]
	learned s1	0.194	0.136	39.64	0.934	−29 % [−30, −27]	−31 % [−33, −29]

Table S2. Complete main comparison at $C = 8$, 60 crops and five draws per crop, with matched nominal dose. Metrics in each row come from the same reconstructions. Changes compare a learned plate with its random initialization under the same prior; brackets are cell-bootstrap 95 % intervals.

Reconstruction	Seed	Crops	W1 change [95 % CI]	W1 improved	Centroid change [95 % CI]
TV	0	12 prespecified	−38 % [−43, −34]	12/12	−37 % [−41, −34]
TV	0	48 random	−42 % [−46, −37]	48/48	−43 % [−49, −38]
TV	1	12 prespecified	−39 % [−42, −36]	12/12	−38 % [−40, −36]
TV	1	48 random	−42 % [−46, −37]	48/48	−44 % [−49, −38]
Pretrained PnP	0	12 prespecified	−44 % [−50, −39]	12/12	−46 % [−53, −41]
Pretrained PnP	0	48 random	−51 % [−54, −47]	48/48	−56 % [−59, −52]
Pretrained PnP	1	12 prespecified	−44 % [−49, −38]	12/12	−46 % [−52, −40]
Pretrained PnP	1	48 random	−51 % [−55, −47]	48/48	−57 % [−60, −52]
Neuron-trained PnP	0	12 prespecified	−27 % [−34, −20]	12/12	−26 % [−35, −18]
Neuron-trained PnP	0	48 random	−28 % [−30, −26]	48/48	−31 % [−34, −28]
Neuron-trained PnP	1	12 prespecified	−29 % [−34, −25]	12/12	−29 % [−35, −24]
Neuron-trained PnP	1	48 random	−28 % [−30, −27]	48/48	−31 % [−34, −29]

Table S3. Separate evaluation on the prespecified and randomly sampled crops. Negative changes favor the learned plate. Both sets contain crops from the same six cells; the 48 additional crops expand spatial coverage rather than biological replication.

crops and the expanded validation grid. Its phase, amplitude and DMD rows share the same 2× aperture, relay, photon model, direction draws and tuning procedure. Learned centroid errors of 0.230–0.234 μm are below all three random phase plates (0.422–0.438 μm) and amplitude masks (0.396–0.523 μm). The learned seed-0 plate reduces centroid error by 42–56 % against the amplitude masks and 45–47 % against the random phase plates. This shows that its initialization is representative of these controls. It does not show that learning succeeds from other initializations.

The two in-focus DMD pattern sets remain close in centroid error and image quality. Their low depth discrimination concerns this fixed-focus control, not all amplitude encoders. Indeed, the defocused amplitude masks recover depth and can match random phase plates in centroid error. The plate-free scan also retains weak finite-aperture diversity, so its observations are not an ideal zero-depth-information system.

Point-emitter diagnostics. For 25 interior lateral locations and 16 depths, we compare measurement columns under a Gaussian approximation of shot noise with unknown emitter brightness. Random phase plates assign the wrong depth in 1.5–1.7 % of cases with 32 directions; amplitude masks give about 5 %, the in-focus DMD about 25 %, and the plate-free control about 89 %. These are model-based emitter diagnostics, not measured resolution of dense structures. The plate-free diagnostic uses the original 2× aperture; its neuronal control in Tab. S4 uses 4×.

Comparisons across priors. A descriptive pairing of learned plate + TV against random plate + neuron-trained PnP gives 19.6 % lower W1 (95 % CI: 16.4–22.5 %), improving 56 of 60 crops. It does not establish a general dominance relation: learned plate + pretrained PnP has 7 % higher mean W1 than random plate + neuron-trained PnP (interval −2.6 to +17.3 %). Optical and prior choices interact, and neuron-trained PnP with the learned plate performs best overall.

Encoder	Centroid (μm)	PSNR (dB)	FG-SSIM
Learned, seed 0	0.230	36.15	0.868
Learned, seed 1	0.234	36.21	0.869
Random phase 0	0.431	34.81	0.784
Random phase 1	0.422	34.93	0.787
Random phase 2	0.438	34.85	0.782
Amplitude mask 0	0.523	33.70	0.703
Amplitude mask 1	0.419	34.37	0.766
Amplitude mask 2	0.396	34.36	0.763
DMD set 0	2.169	28.83	0.482
DMD set 1	2.169	28.81	0.479
Plate-free, 4× aperture	1.102	30.18	0.494

Table S4. Untrained-encoder control study at $C = 8$: 12 prespecified crops, pretrained PnP. W1 was not computed in this study. The plate-free row uses a 4× aperture in a separate study; its own random-plate control gives 0.446 μm/34.82 dB. Other rows use 2× apertures. Do not interpret the plate-free row as an exactly matched cross-aperture contrast.

S6. Structural tasks and measurement discrimination

Crossing phantoms. We build Gaussian fluorescent tubes on the $32 \times 96 \times 96$ analysis grid, at 0.32 μm spacing. Their lateral and axial standard deviations are 0.32 μm, with half-length 12.8 μm. The equal-branch reconstruction phantoms combine tubes by their maximum before $2 \times 2 \times 2$ averaging. Three lateral geometries use filament-a orientation/crossing-angle pairs $(0°, 90°)$, $(30°, 60°)$ and $(60°, 45°)$. Centers are reconstruction planes 7 and 5. Equal-branch gaps are 0, 2, 3, 4, 6 and 8 planes (0, 1.28, 1.92, 2.56, 3.84 and 5.12 μm), giving 36 phantoms, each with five measurement draws. All use $C = 8$, matched nominal dose and parameters selected on the vascular validation crops, not on these phantoms.

Resolved peaks and arm ordering. Peaks are detected on zero-padded axial columns, requiring height and prominence of at least 20 % of that column's maximum. Reference and reconstructed peaks are matched one-to-one within one reconstruction plane, closest first. Crossing columns must carry more than 20 % of each reference filament's maximum column signal and contain reference peaks matched to both planes. The resolved fraction is the share of reference crossing-column signal in columns whose reconstruction has both matched peaks. It is not an absolute recovered-brightness cutoff.

For ordering, each filament's arm columns carry at least 20 % of its own maximum column signal and less than 2 % of the other filament's. Their reconstructed signal-weighted axial centroids must follow the true order, with consecutive arms separated by more than half a plane. A volume succeeds when the resolved fraction is at least 0.5 and ordering is correct. In the zero-gap control, two matched peaks instead count as a false split.

The qualifying gap is the smallest tested gap of at least two planes with success rate at least 0.8 at that gap and every larger tested gap, at both centers separately. Each center's rate uses 15 reconstructions (three geometries and five draws). The main figure pools the centers for its line and shades their range; the criterion uses the individual rates.

The main displayed example uses the first lateral geometry, center 7 and draw 0. Its gap was chosen at the learned-TV threshold, so the gap is outcome-informed. Random and learned TV succeed in 10 % and 90 % of the pooled cases at that gap; the displayed draw is one failed/successful pair. All displayed sections share geometry and intensity scale. Neither plate falsely splits the merged control. The weak-branch and three-filament results in Tab. S5 limit extrapolation beyond the two equal-intensity branches.

Merged neuronal alternatives. A reference column is called layered when it belongs to the signal mask and has at least two peaks under the same relative height/prominence rule. For each corresponding fine-grid lateral column, we move its intensity to its axial centroid, distributing mass between the two neighboring planes to preserve that centroid exactly. Other columns remain unchanged. The resulting alternative preserves the fine-grid lateral projection and each modified column's centroid while changing its depth profile. One crop has no eligible columns, leaving 59. These are constructed candidates, not solutions optimized over a measurement-consistency set; concentrated intensities need not remain below the original normalized peak.

For noise-free expected counts μ of the reference and $\nu(c)$ of a brightness-scaled alternative, we compute

$$D_{\mathrm{P}}(\mu\|\nu) = \sum_i \left[\mu_i \log \frac{\mu_i}{\nu_i} + \nu_i - \mu_i\right]. \qquad \text{(S7)}$$

The simulator evaluates both objects on the fine grid, using each matched direction subset and the same exposure. Without background, the profiled scale is the ratio of total reference to alternative expected counts. With background, the numerical fit searches a log-scale interval of $[-1, 1]$ around the total-signal estimate. Crop scores average five subsets. The learned/random ratio is 9.99 (95 % CI: 8.66–11.79) without background and 10.4 with background. Random-plate divergences can already be large; the ratio is a diagnostic of these specified alternatives, not a probability of successful reconstruction. No KL-to-reconstruction-success guarantee is claimed.

Additional neuronal structure metrics. Layered columns carry about 17 % of reference signal on average. Their W1 decreases by 33.8 % with TV, 42.4 % with pretrained PnP and 27.6 % with neuron-trained PnP. However, the prespecified comparison of profile gain against centroid gain does

	Equal branches		Weak branch		Three filaments	
Reconstruction	Random	Learned	Random	Learned	Random	Learned
TV	3.84	2.56	–	3.84	–	–
Pretrained PnP	3.84	2.56	–	–	–	–
Neuron-trained PnP	3.84	3.84	–	–	–	–

Table S5. Qualifying gaps (μm) under the same success rule. A dash means the criterion is not reached at both centers over the tested gaps. Weak branch: the second filament has 0.3 times the first's intensity. Three-filament cases remain unresolved.

not support a larger profile benefit: for TV the difference is -0.7 percentage points (95 % CI: -2.9 to $+1.5$). Most layered columns are intensity structure within thick branches, so these metrics alone do not establish separated neuronal branches.

The earlier layered-column protocol required each rerun to reproduce its recorded PSNR within 0.05 dB. TV met that check; PnP did not. The PnP layered-column results are therefore descriptive. The later all-column W1 evaluation uses its separately documented numerical check (Sec. S4). We do not relabel the failed earlier gate as a passed structural test.

As a further descriptive measure, centerline Dice [11] compares a reference mask at 10 % of peak intensity with an estimated mask retaining the same number of voxels, followed by 3D skeletonization. Scores rise from 0.79 to 0.84 for TV, 0.75 to 0.84 for pretrained PnP and 0.85 to 0.88 for neuron-trained PnP. This thresholded metric is depth-sensitive but does not substitute for branch tracing or connectivity validation.

S7. Measurement, photon and alignment controls

Dose convention. Each encoder's excitation is divided by its mean over the specimen analysis grid and the full 256-direction scan. This fixes the deposited excitation per complete scan. For k directions, exposure is multiplied by $256/k$. Random subsets therefore match nominal dose in expectation, not exactly for each subset. The per-specimen exposure is set from the random plate's full-scan mean signal and reused for every encoder. Equal deposited dose does not force identical detected counts: illumination can correlate differently with the specimen. For the two learned plates, mean detected signal changes by about -1 to -2 % across the 12 crops, with individual changes approximately -12 to $+9$ %.

Measurement compression. Table S6 reports both depth metrics at $C = 1, 4, 8$. These values count all camera samples, not camera images alone. At matched nominal dose, fewer directions mean longer exposure per image. The learned 32-image acquisition gives lower W1 than the 256-image random acquisition for TV and pretrained PnP, but the close neuron-trained-PnP means do not establish equivalence. The TV curve is not monotone in compression: validation can prefer a stronger regularization weight at $C = 4$, which smooths neuronal features more than at $C = 8$.

Noise conditions. The nominal model uses Poisson shot noise. The background condition adds a uniform mean equal to the random plate's mean expected signal for the same exposure, plus independent Gaussian read noise with standard deviation 2 electrons. The known mean background is subtracted before reconstruction. The same background level, exposure and noise protocol are used for each encoder. A separate low-dose condition reduces exposure to one tenth. Scalar reconstruction parameters are revalidated for each changed noise condition, while networks stay fixed. Table S7 reports all three methods.

Photon budget. The dose sweep uses factors 0.1, 0.3, 1 and 3 of the nominal exposure (Tab. S8). At dose 0.3, learned-plate W1 is 0.234, 0.302 and $0.220\,\mu$m for TV, pretrained PnP and neuron-trained PnP, compared with 0.318, 0.496 and $0.282\,\mu$m for the random plate at dose 1. These reductions of 26 %, 39 % and 22 % describe the tested simulation and inference budgets. The random curve's flattening between doses 1 and 3 is not evidence of a fundamental photon limit.

Alignment. Distance perturbations set $D = 6.1$ or $7.1\,\mu$m. In the characterized case, the operator is rebuilt from the known distance and scalar reconstruction parameters are revalidated; the phase is unchanged. In the nominal-operator case, reconstruction ignores the perturbation. Lateral error shifts the plate by one $0.32\,\mu$m feature. Pointing errors are offsets of 0.005 or 0.01 in transverse direction cosine. All comparisons use neuron-trained PnP, $C = 8$, the same 12 crops and five draws.

Table S9 separates these assumptions. Known distance changes retain the optical gain. Ignoring distance or lateral registration can make the learned plate worse than the random one, including an 11 % W1 increase for a one-feature lateral shift. Rebuilding an operator with a known pertur-

Reconstruction	Plate	W1, $C = 1$	Centroid	W1, $C = 4$	Centroid	W1, $C = 8$	Centroid
TV	Random	0.249	0.187	0.298	0.223	0.318	0.243
TV	Learned, seed 0	0.170	0.135	0.208	0.166	0.196	0.153
TV	Learned, seed 1	0.160	0.124	0.202	0.158	0.195	0.151
Pretrained PnP	Random	0.413	0.357	0.445	0.384	0.496	0.431
Pretrained PnP	Learned, seed 0	0.222	0.184	0.254	0.212	0.277	0.231
Pretrained PnP	Learned, seed 1	0.213	0.173	0.249	0.206	0.279	0.234
Neuron-trained PnP	Random	0.203	0.161	0.234	0.184	0.282	0.221
Neuron-trained PnP	Learned, seed 0	0.142	0.110	0.167	0.130	0.206	0.163

Table S6. Measurement budgets on the 12 prespecified crops. W1 and centroid errors are in μm and come from the same reconstruction rows. $C = 1$ uses all 256 directions; $C = 4, 8$ average five subsets of 64 and 32 directions. The second seed is available across these budgets for TV and pretrained PnP; neuron-trained PnP is evaluated with seed 0 here and both seeds at $C = 8$.

Condition	Reconstruction	W1, random	W1, learned	Centroid, random	Centroid, learned	W1 change	Centroid change
Shot noise	TV	0.318	0.196	0.243	0.153	$-38\,\%$ $[-43, -34]$	$-37\,\%$ $[-41, -34]$
Shot noise	Pretrained PnP	0.496	0.277	0.431	0.231	$-44\,\%$ $[-50, -39]$	$-46\,\%$ $[-53, -41]$
Shot noise	Neuron-trained PnP	0.282	0.206	0.221	0.163	$-27\,\%$ $[-34, -20]$	$-26\,\%$ $[-35, -18]$
Background + read noise	TV	0.324	0.227	0.249	0.180	$-30\,\%$ $[-38, -22]$	$-28\,\%$ $[-36, -20]$
Background + read noise	Pretrained PnP	0.627	0.297	0.548	0.247	$-53\,\%$ $[-57, -48]$	$-55\,\%$ $[-60, -50]$
Background + read noise	Neuron-trained PnP	0.288	0.210	0.228	0.166	$-27\,\%$ $[-35, -19]$	$-27\,\%$ $[-36, -18]$
0.1 dose	TV	0.466	0.292	0.365	0.236	$-37\,\%$ $[-45, -29]$	$-35\,\%$ $[-44, -28]$
0.1 dose	Pretrained PnP	0.834	0.407	0.733	0.352	$-51\,\%$ $[-57, -45]$	$-52\,\%$ $[-58, -45]$
0.1 dose	Neuron-trained PnP	0.345	0.239	0.269	0.186	$-31\,\%$ $[-37, -24]$	$-31\,\%$ $[-38, -24]$

Table S7. Noise controls on 12 crops at $C = 8$. Errors are in μm; changes and cell-bootstrap 95 % intervals compare the learned and random plates within the same condition.

Reconstruction	Nominal dose	W1, random	W1, learned	Centroid, random	Centroid, learned
TV	0.1	0.466	0.292	0.365	0.236
TV	0.3	0.442	0.234	0.341	0.185
TV	1	0.318	0.196	0.243	0.153
TV	3	0.315	0.192	0.242	0.151
Pretrained PnP	0.1	0.834	0.407	0.733	0.352
Pretrained PnP	0.3	0.631	0.302	0.550	0.251
Pretrained PnP	1	0.496	0.277	0.431	0.230
Pretrained PnP	3	0.494	0.276	0.430	0.230
Neuron-trained PnP	0.1	0.345	0.239	0.269	0.186
Neuron-trained PnP	0.3	0.304	0.220	0.236	0.174
Neuron-trained PnP	1	0.282	0.207	0.221	0.164
Neuron-trained PnP	3	0.275	0.203	0.218	0.162

Table S8. Dose sweep at $C = 8$, 12 crops. W1 and centroid errors (μm) are computed from the same reconstructions. Each exposure factor is relative to the matched nominal-dose protocol.

bation is not an experimentally demonstrated calibration procedure. Phase-quantization experiments are not used to claim a fabricable low-level plate: the simulated surround was not quantized with the trained center.

Condition	W1, random	W1, learned	Centroid, random	Centroid, learned	W1 change	Centroid change
none (design)	0.282	0.206	0.221	0.163	−27 % [−34, −20]	−26 % [−35, −18]
Known: D=6.1	0.285	0.205	0.225	0.162	−28 % [−35, −22]	−28 % [−37, −20]
Known: D=7.1	0.284	0.209	0.226	0.166	−27 % [−34, −20]	−26 % [−35, −19]
Nominal operator: scan tilt 0.005	0.318	0.254	0.253	0.196	−20 % [−25, −16]	−22 % [−30, −15]
Nominal operator: scan tilt 0.01	0.379	0.328	0.308	0.251	−13 % [−16, −10]	−18 % [−22, −15]
Nominal operator: D=6.1	0.563	0.586	0.528	0.571	+4 % [+0.3, +8]	+8 % [+3, +14]
Nominal operator: D=7.1	0.447	0.473	0.383	0.438	+6 % [+5, +7]	+14 % [+12, +17]
Nominal operator: lateral shift 1 feature	0.451	0.500	0.372	0.400	+11 % [+2, +23]	+7 % [−0.4, +19]

Table S9. Alignment controls with neuron-trained PnP. Errors are in μm; changes compare learned and random plates under the same perturbation. "Known" means the offset is supplied to operator assembly and validation; "nominal operator" means it is ignored. Positive changes favor the random plate.

S8. Complementary uncertainty analyses

These analyses ask different questions from the main reconstruction tests. Centroid intervals quantify an empirical depth error rate after reconstruction. Set-membership bounds assess admissible voxel intensities without a learned prior. Neither supplies a connectivity guarantee for the neuronal reconstructions.

Conformal centroid intervals. We form $\hat{z}_c \pm \lambda$ around each reconstructed axial centroid. For a volume, $L(\lambda)$ is the reference-signal-weighted fraction of evaluated columns whose true centroid lies outside the interval. It is bounded by one and decreases with λ. Conformal risk control [1] selects

$$\hat{\lambda} = \inf\left\{\lambda : \frac{n}{n+1}\hat{R}_n(\lambda) + \frac{1}{n+1} \le \alpha\right\}, \qquad \text{(S8)}$$

where $\hat{R}_n$ averages the loss over n calibration volumes. Under exchangeability of the calibration volumes and a new volume, $\mathbb{E}[L(\hat{\lambda})] \le \alpha$. This controls expected signal-weighted miscoverage, not each column's or specimen's coverage.

We use neuron-trained PnP at $C = 8$, target $\alpha = 0.1$ and encoder-specific calibration on 35 non-overlapping crops from a held-out SOM009 stack. A 240-crop overlapping set is a sensitivity analysis, not independent replication. On the 12 prespecified test crops, the learned seed-0 half-width decreases from 0.646 to 0.474 μm (26.7 %), with empirical miscoverage 0.075 versus 0.060 for the random plate (Tab. S10). The test animals differ from the calibration animal, so exchangeability is not established; observed miscoverage is empirical rather than a cross-animal guarantee.

Prespecified connectivity calls based on reconstructed bottlenecks did not meet their error target: false-disconnected rates were 0.23–0.48 against a 0.15 limit, with no connected test call. We therefore do not convert centroid calibration into error-controlled connectivity statements.

Voxel bounds without a learned prior. Given $\mathbf{x} \in [0,1]^N$, define

$$\mathcal{C}(\mathbf{y}) = \{\mathbf{x} : \ell_i \le (\mathsf{A}\mathbf{x})_i \le h_i,\ 0 \le \mathbf{x} \le 1\}. \qquad \text{(S9)}$$

	35 non-overlapping crops		all 240 crops	
plate	$\hat{\lambda}$ (μm)	test miscov.	$\hat{\lambda}$ (μm)	test miscov.
random	0.646	0.060	0.570	0.085
learned, seed 0	0.474	0.075	0.422	0.091
learned, seed 1	0.486	0.060	0.422	0.081

Table S10. Calibrated centroid half-widths and empirical signal-weighted test miscoverage. The 240-crop analysis reuses overlapping regions. Seed 1 gives 0.486 μm on the primary 35-crop calibration set.

For a nonempty set, minimizing and maximizing x_j gives the smallest and largest voxel intensities compatible with the stated constraints [5, 8]. We solve these linear programs with HiGHS [6]. The reconstruction's quadratic data term does not automatically place its output in this set.

The reduced field contains $4 \times 12 \times 12$ reconstruction voxels and 3×3 camera samples per image, using 32 fixed directions. Thus, $N = 576$ and $M = 288$; this is a separate geometry, not the full-field $C = 8$ experiment. Plates are trained separately on this reduced geometry; no reflected-depth pair qualifies for the full-grid weighting rule, so all off-diagonal weights are one.

The ancillary Gaussian model uses reference full-scan mean m_p for the random plate and mean m_e for an encoder, with $\sigma_e = m_p 10^{-q/10}\sqrt{k/256}\sqrt{m_e/m_p}$ and reference parameter $q = 15$. At $q = 15$, this is a spatially uniform Gaussian approximation to the main photon model at each encoder's mean signal, rather than a pixel-wise Poisson variance. The formula defines q explicitly; it is not a standard RMS SNR. Intervals use $y_i \pm z\sigma_e$, where $z = \Phi^{-1}(1 - 0.05/(2 \cdot 2880))$. The family contains all 2304 rows of the 256-direction reduced scan plus up to 576 projection constraints. Using the same correction for every subset gives at least 95 % simultaneous coverage under the stated Gaussian model and exact reduced linear operator. Unmodeled optical or specimen discretization error would require a separate allowance.

Across 24 synthetic phantoms and three noise realizations

each, the learned reduced-field plates narrow mean voxel-wise bound width by about 21 %. On 48 neuronal windows, however, bounds span the full admissible range for every voxel at the tested settings. Better neuronal reconstruction in the main study therefore does not imply unique voxel intensities or uniquely identified connections. These negative neuronal results are retained as limits of the measurement-only analysis, not as optical-learning performance measures.

S9. Neuronal views and display selection

Reference-only selection. Before opening reconstructions for figure selection, we computed reference features for all 60 crops. Eligible crops had at least 10 skeleton endpoints (the upper quartile) and a 10–90 % column-centroid range of at least $3.64\,\mu\mathrm{m}$ (the median), indicating several branches and substantial depth extent. Eight crops qualified. We ranked them by reference signal in columns with at least two axial peaks. The first was SOM004_cell2_x7; the companion was the highest-ranked eligible crop from another animal, SOM006_cell1_x7.

After selection, their W1-gain ranks among 60 crops were 52/45/23 and 45/46/43 for TV/pretrained/neuron-trained PnP, where rank 1 is the largest reduction. These are typical-to-lower-gain examples. The opening figure displays pretrained PnP, the prior with the largest cohort-average gain; Fig. S1 shows all priors for that same crop. No larger reconstructed field exists in this evaluation.

Views and scales. The axial band maximizes reference signal in layered columns over eight consecutive lateral rows ($5.12\,\mu\mathrm{m}$). Its lateral window spans 24 voxels ($15.36\,\mu\mathrm{m}$), centered on that reference signal. Lateral context is the full maximum-intensity projection; axial views are maxima over the selected band with true physical aspect ratio and depth increasing downward. Linear display limits are zero and the reference projection's 99.5th percentile, shared across the matched panels of each row. There is no gamma adjustment, smoothing or per-panel rescaling. Error maps use the reference signal mask and a common 0–2 $\mu\mathrm{m}$ W1 scale. All displayed volumes use draw 0.

A feature missed by both plates. We select one column in the companion band using reference signal alone: the layered column with largest reference mass whose two highest peaks are at least three planes apart. At $(x, y) = (18, 3)$, a weak shallow peak at plane 2 lies above a deeper peak at plane 9. The shallow peak is 24.5 % of the column maximum and carries 7.8 % of its signal in planes 1–3. None of the six reconstructions has a matched shallow peak; that signal becomes a low plateau (Fig. S5). This limit coexists with lower average W1 and improved placement of brighter features.

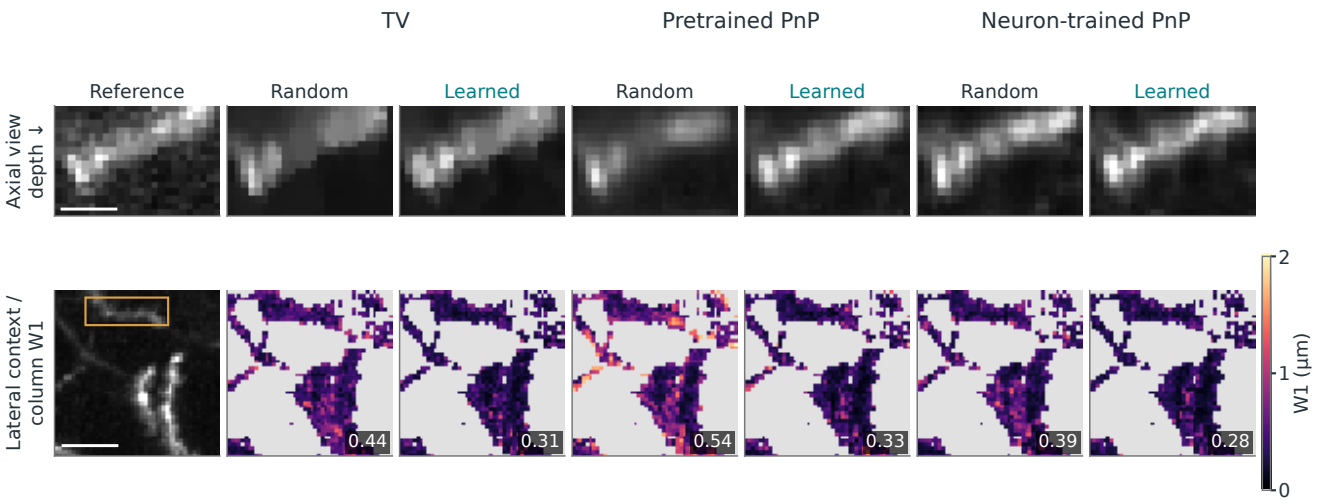


Figure S1. All three priors for the opening figure's reference-selected crop. Top: matched axial band views; bottom: reference lateral context and column-W1 maps. Gray is outside the reference signal mask. Panel values are signal-weighted crop W1 for this draw, in μm. Scale bars: $5\,\mu$m in band views and $10\,\mu$m in lateral context.

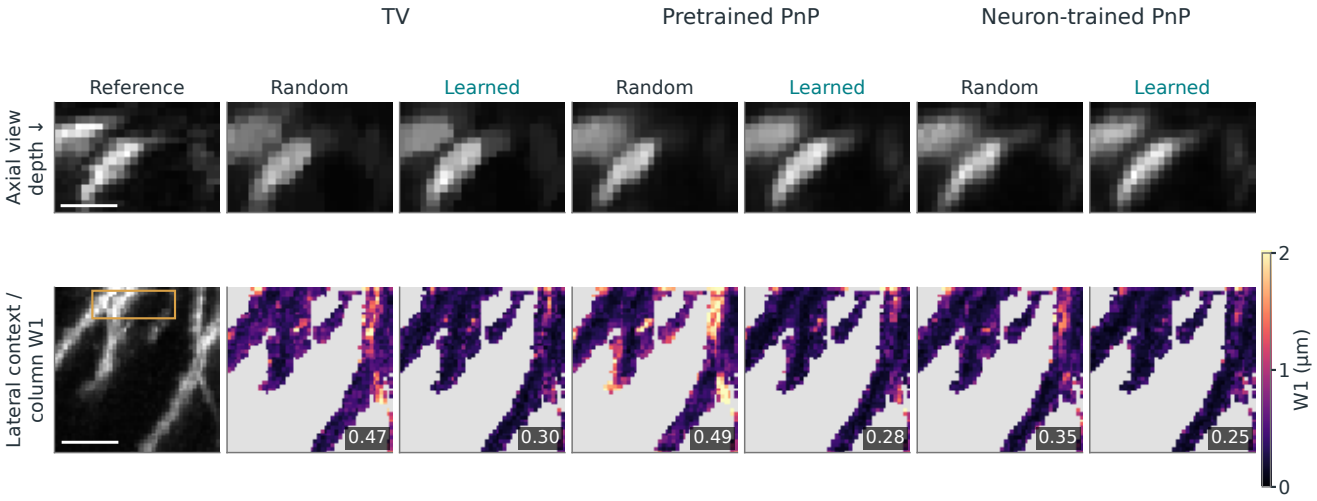


Figure S2. The reference-selected companion from another animal. Top: matched axial band views; bottom: reference lateral context and column-wise W1 maps. Panel values are draw-0 signal-weighted crop W1 in μm. Shared linear scales and view conventions match the first crop. Bars: $5\,\mu$m in band views and $10\,\mu$m in lateral context.

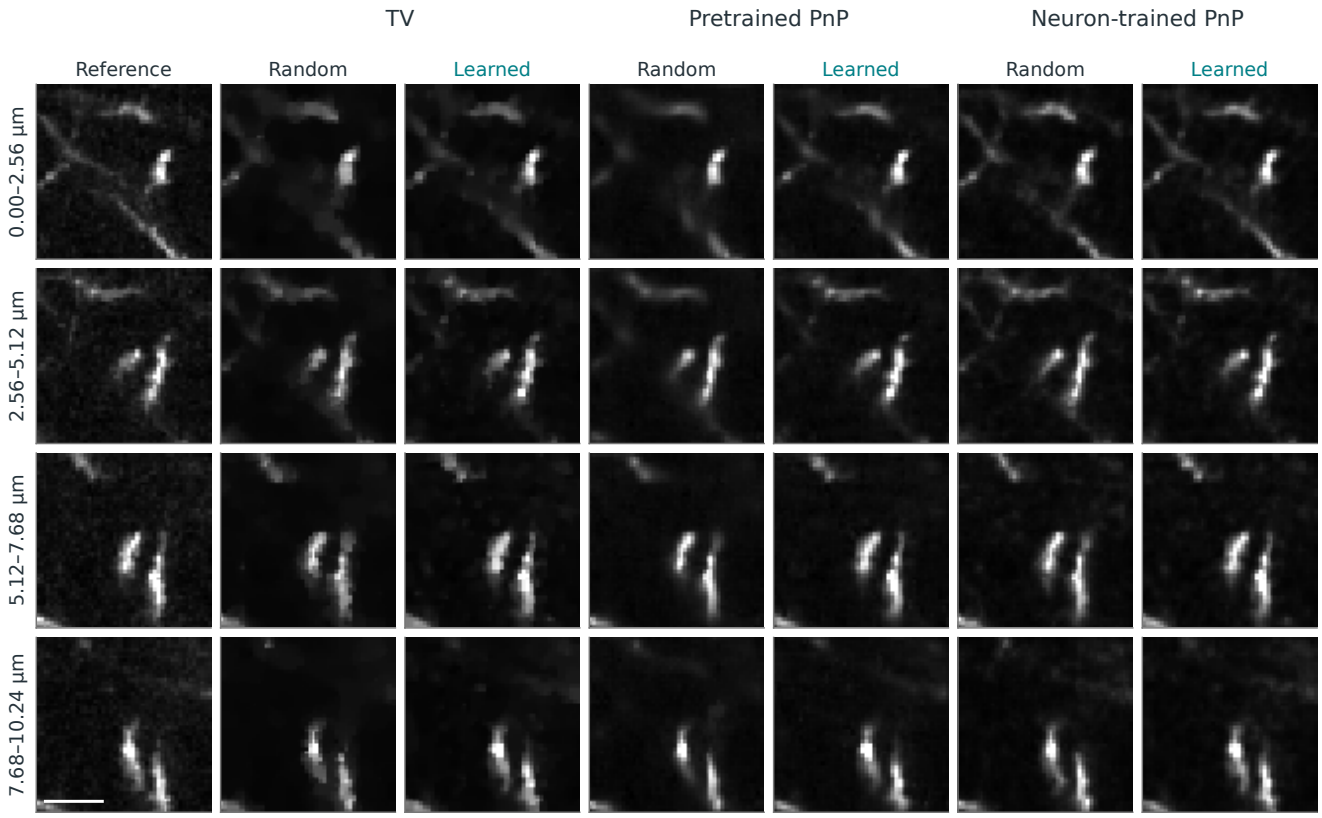


Figure S3. Depth slabs of the opening neuronal example, with maximum projections over four consecutive depth planes ($2.56\ \mu$m per slab). Reference and all reconstructions share the same linear intensity range across slabs. Slab views expose signal spread into depths absent from the corresponding reference slab. These are qualitative placement comparisons, not branch-separation measurements. Bar: $10\ \mu$m.

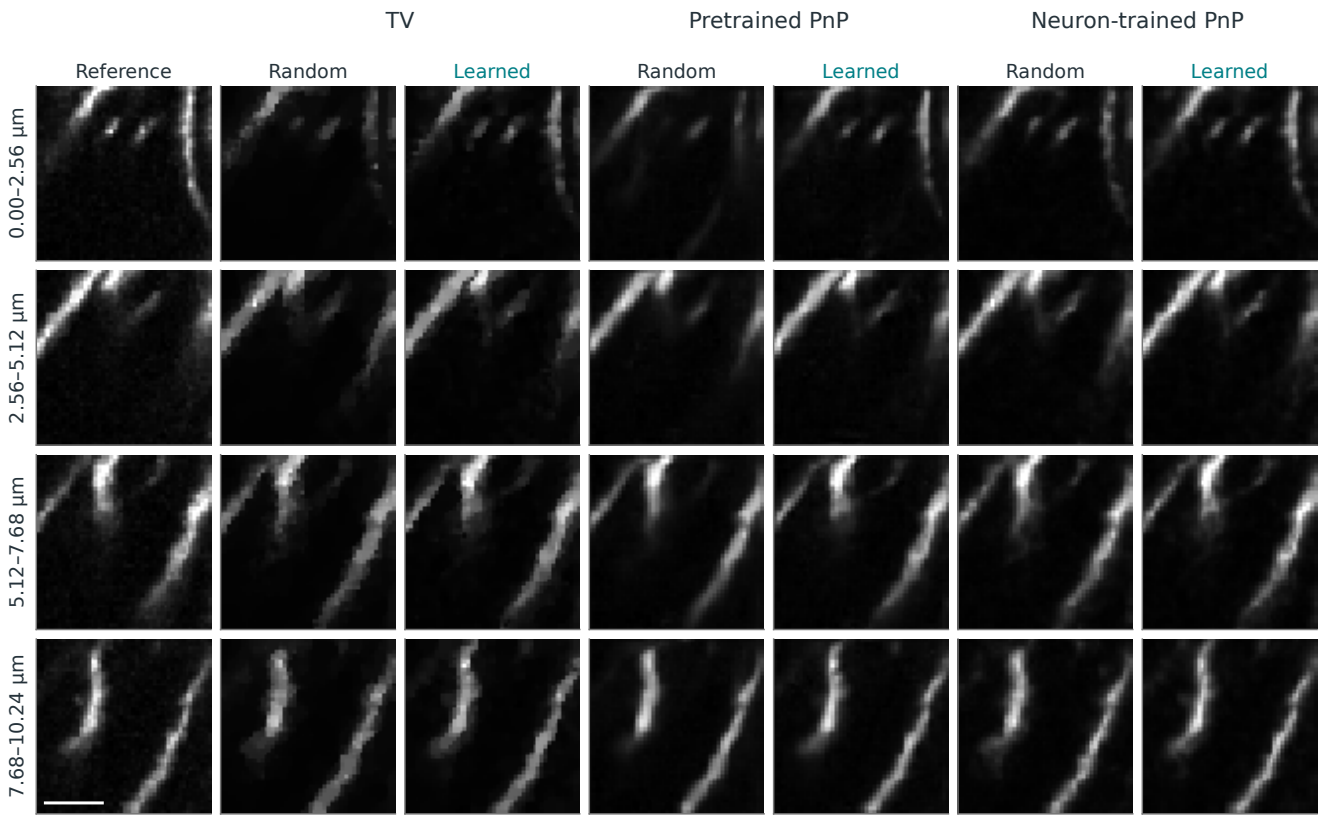


Figure S4. The companion crop from another animal, with the same depth-slab conventions. Stronger priors reduce diffuse signal for both plates; the optical difference is smaller under neuron-trained PnP. The full views retain features missed by either encoder.

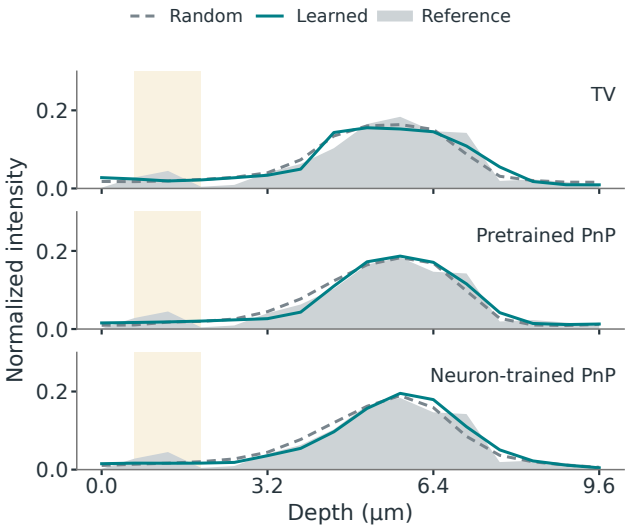


Figure S5. Reference-selected weak-feature column in the companion crop. Profiles are normalized to unit mass. Both plates miss the shallow reference peak under all three priors, despite the lower average W1 of the learned plate. This column was fixed from reference structure before inspecting reconstructions.

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